

Exposing Deepfake Videos by Tracking Eye Movements

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Abstract—It has recently become a major threat to the public media that fake videos are rapidly spreading over the Internet. The advent of Deepfake, a deep-learning based toolkit, has facilitated a massive abuse of improper synthesized videos, which may influence the media credibility and human rights. A worldwide alert has been set off that finding ways to detect such fake videos is not only crucial but also urgent. This paper reports a novel approach to expose deepfake videos. We found that most fake videos are markedly different from the real ones in the way the eyes move. We are thus motivated to define four features that could well capture such differences. The features are then fed to SVM for classification. It is shown to be a promising approach that without high dimensional features and complicated neural networks, we are able to achieve competitive results on several public datasets. Moreover, the proposed features could well participate with other existing methods in the confrontation with deepfakes.

I. INTRODUCTION

For a long time, synthesizing images or videos has been an interest of both academics and film making industry. From the perspective of synthesizing methods, early fake images or videos were mainly manufactured with graphic algorithms. And it has not grabbed the public attention until recently when a technology known as Deepfake [1], [2] was made openly accessible. With Deepfake, which is intensively dependent on deep neural networks, fabricating a video clip of people doing or saying things by design involves nothing more than a few clicks. Without doubt, the impact of deepfake videos has stretched rapidly far beyond what the creators expected, which may affect how people distinguish authentic media information. The belief of “seeing is believing” has been shaken. While it is certainly innocuous when the technology is applied to enriching our entertainment lives, serious consequences may be caused given that such videos may be “weaponized” by some people for political or other malicious purposes.

In response to this situation, researchers have begun to explore various methods to distinguish deepfake videos from authentic ones [3], [4]. Most of the detection methods are based on machine learning. While some works rely on neural networks of different architectures to discover the disparities between real and fake videos including [5]–[8], others try to leverage handcraft features that are semantically meaningful, such as the pattern of eye blinking [9], head poses [10], visual artifacts [11], face warping traces [12], the individual habit of facial expressions and movements when speaking [13],

etc. To our best knowledge, most methods either just rely on features of individual frames, ignoring temporal information, or depend too heavily on the training datasets, and thus lack generalization. Generally speaking, current research on deepfake video forensics is still in its infancy.

In this paper, taking both temporal and spatial information into consideration, we make use of physiology information to discriminate between real and fake videos. Inspired by the synthesis mechanism of deepfake videos and the characteristic of human eye movements, we explore the pattern of eye movements and apply it to exposing deepfake videos. Specifically, we defined four features about the binocular movements which are fed to SVM for classification. Intuitive as it is, our proposed approach proves to be quite effective, which could be seen as a reinforcement to the counterattack against deepfakes.

II. PROPOSED METHOD

A. Motivation and Pipeline of the Method

Essentially, the Deepfake toolkit works by altering the critical features of the face while keeping most background and insignificant features untouched. With subtle alteration, the character of the video has been changed. In doing so, the message conveyed by the video could be completely different. Such alteration is conducted on a frame-by-frame basis, which is believed to be the major limitation of Deepfake. Although it is possible for a well-designed algorithm to fine-tune each individual image and make it lifelike, it is very hard to coordinate the reality of a whole set of images on which various temporal constraints are imposed.

Most deepfake videos reflect a series of movements and reactions of a person. As long as the person is physically and mentally healthy, his/her facial expressions and body movements tend to exhibit an intrinsic consistency. Take the eyes for example, when we talk and think, it is natural for us to look around a little bit or to change our vision focus from time to time. Such eye-gaze movements are spontaneous and consistent. Eye movement is a kind of complicated motion controlled by three pairs of extraocular muscles [14]. Different movements such as saccade and smooth tracking have different speed and duration. The left and right eyes of a person are less likely to move in opposite directions simultaneously,

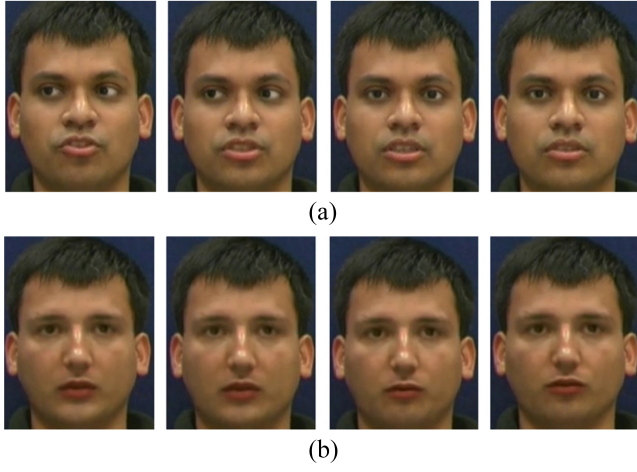


Fig. 1. Examples of corresponding frames of a real video and the deepfake counterpart. (a) Frame 40, 43, 46 and 49 of the real video. (b) Frame 40, 43, 46 and 49 of the deepfake counterpart.

which is referred to as the binocular coordination linkage [15]. There is also a limitation on how fast the eyeballs move. All these details work naturally on a human being but are quite difficult for a synthesized character to mimic. Indeed, we have observed from a wide range of deepfake videos that most of them have hardly given any consideration to eye movements. One example from Deepfake-TIMIT dataset [16] is shown in Fig.1. It can be seen that eyes on the real video frames move quickly while the eyes on its fake ones barely move. We are thus motivated to derive a set of features to characterize how realistically the eyes move and based on which, we are able to expose fake videos with high accuracy.

The flowchart of our method is shown in Fig.2. It consists of two stages: gaze estimation stage and manipulation detection stage.

B. Gaze Estimation Stage

In order to track the eye movements of a video clip, we first estimate the gaze orientations of each frame. There are two kinds of gaze estimation methods, i.e. appearance-based [17] and model-based [18]. Appearance-based methods make use of machine learning algorithms to establish the mapping relationship between eye images and gaze orientations directly, which requires a big amount of user-specific eye images for training [19]. Model-based methods consider the eyeball as a geometric sphere model, with which one can infer the gaze orientations. Obviously, there are not adequate training images of user-specific eye images in most of the deepfake detection applications, we thus employ the model-based approach in this work.

We first detect facial landmarks including the eyelids, the iris and the pupil [20], with which we estimate the center of the eyeball. We then compute the gaze vector pointing from the eyeball center to the pupil [21]. A simplified eyeball model is shown in Fig.3 for illustration, where the gaze orientation is denoted as $\mathbf{g} = (g_x, g_y, g_z)$.

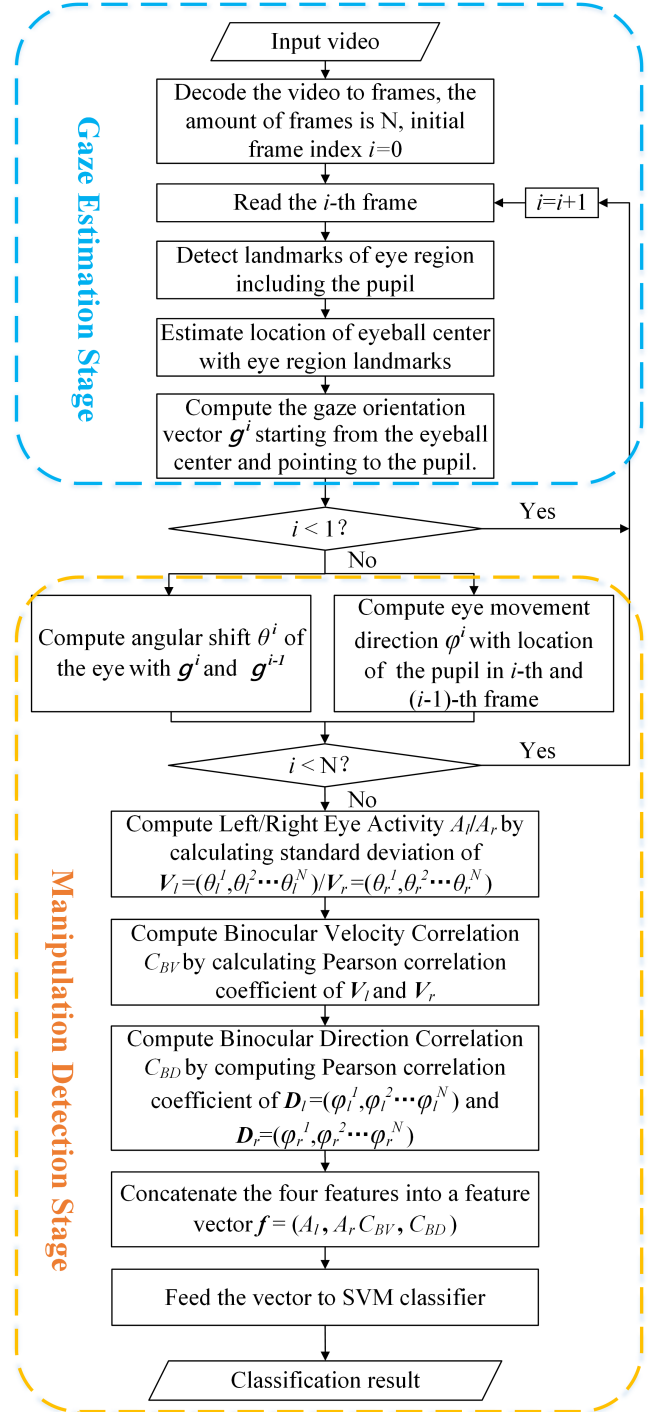


Fig. 2. Pipeline of our method.

C. Manipulation Detection Stage

Our detection approach is based on two observations. First, many fake characters in deepfake videos have glazed eyes, with significantly less eye movement than real human beings. Second, unlike real eyes, fake eyes do not necessarily show the binocular coordination linkage. To reflect such observations, we define the following four measurements to characterize the

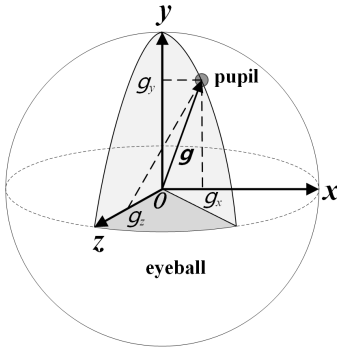


Fig. 3. Illustration of the eye model.

difference of eye movements between real videos and fake ones, namely the left eye activity, the right eye activity, the binocular angular shift correlation and the binocular direction correlation.

1) *Left eye activity and right eye activity:* Let vectors $\mathbf{g}^i = (g_x^i, g_y^i, g_z^i)$ and $\mathbf{g}^{i-1} = (g_x^{i-1}, g_y^{i-1}, g_z^{i-1})$ denote the gaze orientations in two successive frames, respectively. The angular shift θ^i between the i th frame and the $(i-1)$ th frame is shown in Fig.4(a) and can be determined as follows:

$$\theta^i = \arccos \left(\frac{\mathbf{g}^i \cdot \mathbf{g}^{i-1}}{\|\mathbf{g}^i\| \cdot \|\mathbf{g}^{i-1}\|} \right) \quad (1)$$

A large angular shift indicates rapid eye movement. Based on this, we introduce a new measurement to describe the eye activity. The eyes activity can reflect the status of the eyes in a video. For an N -frame video clip, let $\mathbf{V}_l = (\theta_l^1, \theta_l^2, \dots, \theta_l^{N-1})$ and $\mathbf{V}_r = (\theta_r^1, \theta_r^2, \dots, \theta_r^{N-1})$ denote the angular shift sequences of the left and right eyes for the video clip, respectively. Then, the left eye activity can be calculated by:

$$A_l = \sqrt{\frac{\sum_{i=1}^{N-1} (\theta_l^i - \bar{\theta}_l)^2}{N-1}} \quad (2)$$

Where $\bar{\theta}_l$ is the average of all the θ_l^i . Similarly, we can calculate the eye activity A_r for the right eye. For real eyes, the values of both A_l and A_r are greater than their corresponding ones for fake eyes.

2) *Binocular angular shift correlation:* We introduce the binocular angular shift correlation to describe whether the left and right eyes shift their angles in the same way. The matching of the angular shifts of the left and right eyes is naturally true for real human beings in a video but is not easily mimicked by a synthetic face video. The angular shift similarity between the two eyes for an N -frame video clip can be calculated by:

$$C_{BAS} = \text{corr}(\mathbf{V}_l, \mathbf{V}_r) \quad (3)$$

$$\text{corr}(\mathbf{V}_l, \mathbf{V}_r) = \frac{\sum_{i=0}^{N-1} (\theta_l^i - \bar{\theta}_l) (\theta_r^i - \bar{\theta}_r)}{\sqrt{\sum_{i=0}^{N-1} (\theta_l^i - \bar{\theta}_l)^2} \sqrt{\sum_{i=0}^{N-1} (\theta_r^i - \bar{\theta}_r)^2}} \quad (4)$$

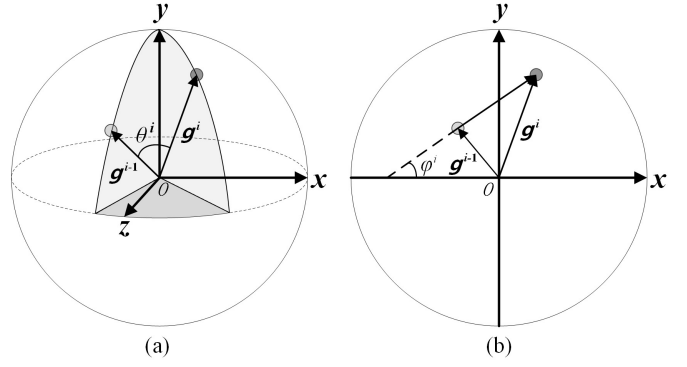


Fig. 4. Illustration of gaze movement. (a) the angle θ between two gaze vectors in two successive frames. (b) the gaze movement direction φ .

where the operator $\text{corr}(\cdot, \cdot)$ calculates the Pearson correlation coefficient of these two sequences. For real eyes, the value of C_{BAS} is greater than that of fake ones.

3) *Binocular direction correlation:* We introduce the binocular direction correlation to describe whether the left and right eyes move in the same direction. Since human eyes rarely move backwards and forth, we only consider the moving direction on the X-Y plane and define the moving direction projection φ^i as shown in Fig.4(b). The moving direction φ^i can be calculated as follows:

$$\varphi^i = \begin{cases} \arccos \frac{g_x^i - g_x^{i-1}}{\sqrt{(g_x^i - g_x^{i-1})^2 + (g_y^i - g_y^{i-1})^2}}, & g_y^i - g_y^{i-1} > 0 \\ -\arccos \frac{g_x^i - g_x^{i-1}}{\sqrt{(g_x^i - g_x^{i-1})^2 + (g_y^i - g_y^{i-1})^2}}, & g_y^i - g_y^{i-1} < 0 \\ \pi/2 - \pi/2 \times \text{sgn}(g_x^i - g_x^{i-1}), & g_y^i - g_y^{i-1} = 0 \end{cases} \quad (5)$$

with

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases} \quad (6)$$

For an N -frame video clip, we compute the binocular direction correlation by:

$$C_{BD} = \text{corr}(\mathbf{D}_l, \mathbf{D}_r) \quad (7)$$

$$\text{corr}(\mathbf{V}_l, \mathbf{V}_r) = \frac{\sum_{i=0}^{N-1} (\varphi_l^i - \bar{\varphi}_l) (\varphi_r^i - \bar{\varphi}_r)}{\sqrt{\sum_{i=0}^{N-1} (\varphi_l^i - \bar{\varphi}_l)^2} \sqrt{\sum_{i=0}^{N-1} (\varphi_r^i - \bar{\varphi}_r)^2}} \quad (8)$$

where $\mathbf{D}_l = (\varphi_l^1, \varphi_l^2, \dots, \varphi_l^{N-1})$ and $\mathbf{D}_r = (\varphi_r^1, \varphi_r^2, \dots, \varphi_r^{N-1})$ denote the moving direction sequences of the left and right eyes for the video clip, respectively. It is expected that these two sequences would have a higher correlation value for real eyes.

Having defined the above four distinctive features, we can construct the feature vector $\mathbf{f} = (A_l, A_r, C_{BAS}, C_{BD})$ for

distinguishing between the real and fake videos. The feature vector is fed to support vector machine (SVM) for classification.

III. EXPERIMENTS

We conducted experiments on a Windows 7 system with a CPU Intel(R) Core(TM) i7-6700@3.4GHz and 12G memory. We tested our approach on the videos from three publicly available datasets, the low quality (LQ) and the high quality (HQ) parts of the Deepfake-TIMIT dataset [16] and the FaceForensics++ dataset [22]. Since the fake videos are generated with different methods in the datasets, we only tested on those fake ones made with Deepfake tools for our evaluation. We first demonstrate the effectiveness of the individual feature through one sample clip and then evaluate the performance of the concatenated feature vector on the three datasets.

A. Feature Effectiveness

V_l and V_r reflect how rapidly the left and right eyes move within a period of time. Fig.5 and Fig.6 show the comparison of V_l and V_r respectively between a real video (ori-sx199-mpgl0.avi from the Deepfake-TIMIT dataset) and its fake counterpart (sx199-video-mwbt0.avi). Example frames of the two videos are given in Fig.1. It can be seen that the curves of the real video exhibit noticeably more fluctuation than those of the fake ones. This indicates that the fake eyes are indeed less vivid than the real eyes. For space limitation, we show only a small segment (from the 50th to the 90th frame) of one pair of example for demonstration. We have actually done a lot of experiments and most of the observations are similar with this example.

D_l and D_r denote the moving directions of eyes in a video. By plotting the two series together we can tell whether the left and the right eyes are moving towards the same directions. As can be seen from the upper part of Fig.7, the two curves of the real eyes are changing consistently at most of the time. However, there is an obvious discrepancy in the two curves of the fake eyes, with quite a few times when the left and right eyes are moving at completely opposite directions.

We then calculate the four features A_l , A_r , C_{BAS} and C_{BD} of the two videos, which are shown in Table I. The left eye activity A_l of the real video is as five times as the value of the fake video. The binocular angular shift correlation C_{BAS} of the real video is as nearly eight times as that of the fake video. To further prove the effectiveness of the proposed features above, in Fig.8 we plot the scatter diagram of 500 randomly selected videos from the Deepfake-TIMIT dataset in feature space consisting of A_l , C_{BAS} and C_{BD} . It shows that the real and fake videos are well separated apart with the proposed features.

B. Statistical Result

In order to test the overall performance of our method, we conduct experiments on the three datasets mentioned. Table II illustrates the division for training and testing sets in our

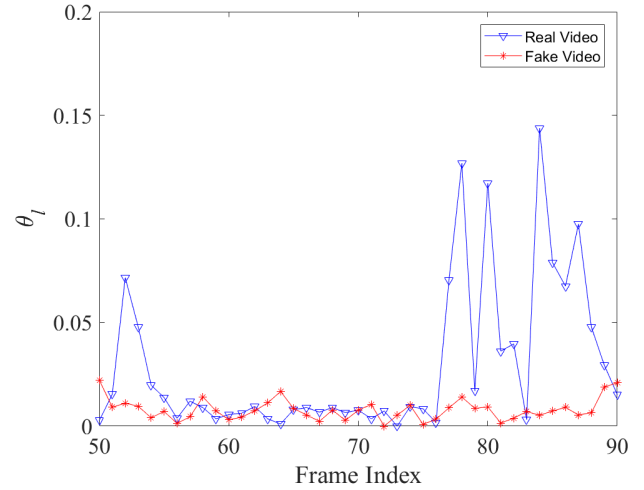


Fig. 5. V_l curves of real and fake videos.

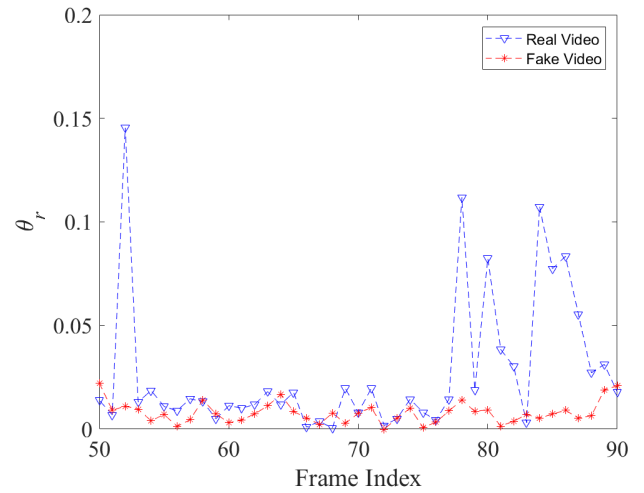


Fig. 6. V_r curves of real and fake videos.

TABLE I
FEATURES VALUES COMPARISON FOR THE REAL AND FAKE SAMPLES.

video name	type	A_l	A_r	C_{BAS}	C_{BD}
ori-sx199-mpgl0.avi	real	0.0250	0.0274	0.85	0.49
sx199-video-mwbt0.avi	fake	0.0050	0.0067	0.11	0.27

TABLE II
THE NUMBER OF TRAINING AND TESTING VIDEOS USED FOR EVALUATION.

Dataset	Num. of Training Videos (Pos. / Neg.)	Num. of Testing Videos (Pos. / Neg.)
FaceForensics++	800	240
Deepfake	(400/400)	(120/120)
Deepfake-TIMIT HQ	300	200
	(150/150)	(100/100)
Deepfake-TIMIT LQ	300	200
	(150/150)	(100/100)

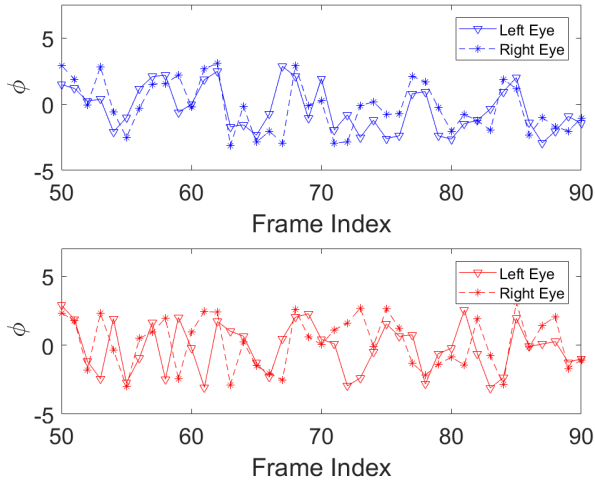


Fig. 7. D_l and D_r curves for real (up) and fake (down) videos.

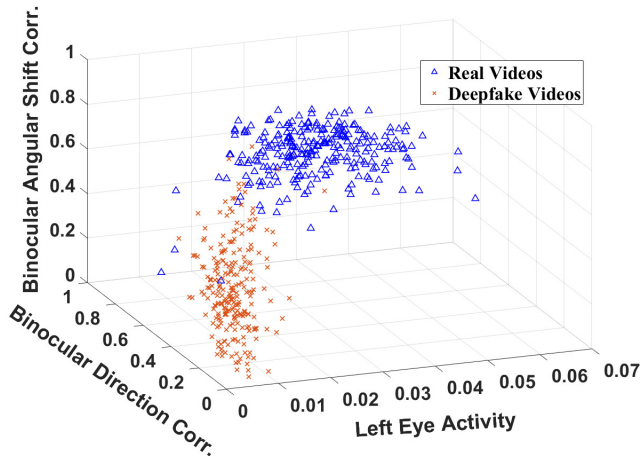


Fig. 8. The relation among the binocular angular shift correlations, the binocular direction correlations and the left eye activities of 500 videos.

experiments. We have ensured the balance between genuine and fake samples in all settings.

The performance of our method is demonstrated with the ROC curves shown in Fig.9. The AUC (area under ROC) for the three datasets are listed in Table III. We also report the EER (Equal Error Rate) and the TPR (True Positive Rate) when FPR (False Positive Rate) equals 10%.

Among the three datasets, our approach works best on the Deepfake-TIMIT LQ set. The results obtained on the other two datasets are also satisfactory though not as good as that on Deepfake-TIMIT LQ. It is believed that the deepfake videos in the Deepfake-TIMIT HQ dataset, due to the higher resolution of input facial images, have a better quality of fabricating than that of the Deepfake-TIMIT LQ, making it more difficult to detect. However, the situation of FaceForensics++-Deepfake dataset is a bit different. The fabricating quality of this set is actually lower with a typical low resolution of the eyes. To our knowledge, the pristine videos in the FaceForensics++-

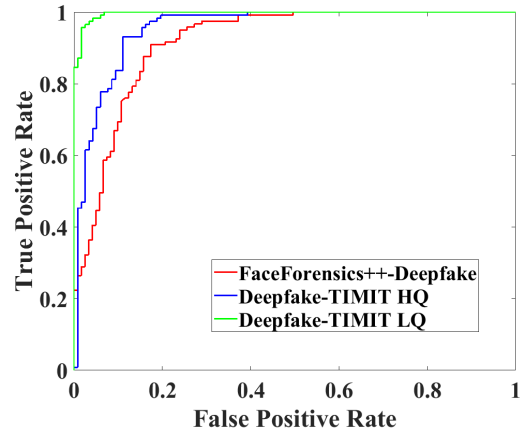


Fig. 9. ROC curves of the SVM classification results on three datasets.

TABLE III
PERFORMANCE OF CLASSIFICATION ON THREE DATASETS.

Dataset	AUC	EER	TPR@FPR=0.1
FaceForensics++-Deepfake	0.918	0.157	0.752
Deepfake-TIMIT HQ	0.955	0.110	0.840
Deepfake-TIMIT LQ	0.996	0.030	1

Deepfake dataset are collected from YouTube, where videos were recorded under multifarious circumstances. As a result, there are a number of videos with blurred eye areas in the dataset, which makes it difficult to track gaze orientations precisely. That explains why the result on the FaceForensics++-Deepfake dataset is not as good as expected.

So far there has not been a benchmark in the field of deepfake video detection yet. The strategies and datasets employed for evaluation vary from one paper to another. It is thus difficult to provide a fair comparison. However, we could list a few relative results compared with our method in Table IV for readers' reference. Despite the fact of different experiment protocols, our detection approach provides a competitive result. This is because other methods extract features on a frame basis without taking temporal motion information into consideration. Since the Deepfake toolkits are adept at fabricating high-quality images, it is not easy to distinguish between real and fake videos with a single frame. As for our method, we take both temporal and spatial information into account, leading to a better performance.

To further demonstrate the generality of our method, we also did cross-database test, which is shown in Table V. When we use the FaceForensics++-Deepfake dataset as the training set, the results on the Deepfake-TIMIT HQ and LQ are nearly as good as that of the intra-dataset test. When we use the Deepfake-TIMIT HQ or LQ dataset as the training set, the result on the FaceForensics++-Deepfake dataset has a certain decrease but is still acceptable.

TABLE IV
AUC VALUES COMPARED TO OTHER METHODS.

Method	FaceForensics++-Deepfake	Deepfake-TIMIT HQ	Deepfake-TIMIT LQ
[7]	0.701	0.735	0.835
[10]	0.473	0.532	0.551
[11]	0.780	0.773	0.770
[12]	0.792	0.932	0.999
proposed	0.918	0.955	0.996

TABLE V
CROSS-DATABASE TEST RESULT

Training database	Testing database	AUC
Deepfake-TIMIT LQ	Deepfake-TIMIT HQ	0.951
	FaceForensics++-Deepfake	0.788
Deepfake-TIMIT HQ	Deepfake-TIMIT LQ	0.994
	FaceForensics++-Deepfake	0.825
FaceForensics++-Deepfake	Deepfake-TIMIT LQ	0.986
	Deepfake-TIMIT HQ	0.945

IV. CONCLUSION

We propose a novel approach for detecting fake videos synthesized with Deepfake tools from the perspective of physiology. Taking both temporal and spatial information into consideration, we have defined four features derived from eye movements to reveal the activity and coordination of eyes in the videos, which can be exploited to tell apart fake and genuine videos. Our approach works with a very small number of features and does not involve complicated neural networks. The results obtained on three third-party datasets proved its effectiveness. It is believed that the performance of our method could be further boosted with the development of more advanced gaze tracking technique. For example, we may first roughly estimate the type of eye movement like saccade or smooth tracking and then detect whether the motion curve comply with the specific pattern of that type of eye movement. By doing this, we may achieve more accurate detection. In addition, the proposed features could readily be incorporated into a comprehensive detection scheme that assembles various types of features, which we believe is a promising future work.

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